
$$g = 9.8 \text{ m/s}^2 \quad G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2 \quad M_E = 5.97 \times 10^{24} \text{ kg} \quad R_E = 6370 \text{ km}$$

$\mathbf{v} = \frac{d\mathbf{r}}{dt}$	$I = \sum_i m_i r_i^2$
$\mathbf{a} = \frac{d\mathbf{v}}{dt}$	$K_{\text{rot}} = \frac{1}{2} I \omega^2$
$v_x = v_{x0} + a_x t$	$ \mathbf{A} \times \mathbf{B} = AB \sin \theta$
$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	$\hat{i} \times \hat{j} = \hat{k}$
$v_x^2 = v_{x0}^2 + 2a_x (x - x_0)$	$\mathbf{L} = \mathbf{r} \times \mathbf{p} = I\vec{\omega}$
$R = v_0^2 \sin(2\theta)/g$	$\vec{\tau}_{\text{net}} = I\vec{\alpha} = \frac{d\mathbf{L}}{dt}$
$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$	$\vec{\tau} = \mathbf{r} \times \mathbf{F}$
$\mathbf{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$	$I_{\text{hoop}} = MR^2$
$\mathbf{r} = x\hat{i} + y\hat{j} + z\hat{k}$	$I_{\text{disk}} = \frac{1}{2} MR^2$
$a = \frac{v^2}{r}$	$I_{\text{sphere}} = \frac{2}{5} MR^2$
$\mathbf{F}_{\text{net}} = m\mathbf{a}$	$I_{\text{rod, middle}} = \frac{1}{12} ML^2$
$F_{\text{fr}} \leq \mu_s F_N$	$I_{\text{rod, end}} = \frac{1}{3} ML^2$
$F_{\text{fr}} = \mu_k F_N$	$\omega = \omega_0 + \alpha t$
$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$	$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$
$W = \mathbf{F} \cdot \Delta \mathbf{r}$	$\omega^2 = \omega_0^2 + 2\alpha (\theta - \theta_0)$
$K = \frac{1}{2} mv^2$	$\omega = \frac{d\theta}{dt}$
$W_{\text{net}} = K_f - K_i$	$T = \frac{1}{f} = \frac{2\pi}{\omega}$
$P = \mathbf{F} \cdot \mathbf{v}$	$\omega^2 = \frac{k}{m}$
$F_G = mg$	$\omega^2 = \frac{g}{l}$
$F_{\text{sx}} = -kx$	$x = A \cos(\omega t + \phi)$
$U_G = mgy$	$x = Ae^{-\frac{b}{2m}t} \cos(\omega t + \phi)$
$U_S = \frac{1}{2} kx^2$	$E = \frac{1}{2} kA^2$
$\mathbf{p} = m\mathbf{v}$	$F = G \frac{m_1 m_2}{r^2}$
$\mathbf{J} = \mathbf{F}_{\text{av}} \Delta t = \Delta \mathbf{p}$	$U = -G \frac{m_1 m_2}{r}$
$(v_{1i} - v_{2i}) = -(v_{1f} - v_{2f})$	$\frac{T^2}{r^3} = \text{constant}$
$\mathbf{r}_{\text{cm}} = \frac{(\sum_{i=1}^n m_i \mathbf{r}_i)}{(\sum_{i=1}^n m_i)}$	
$v = r\omega$	
$a_r = \omega^2 r$	
$a_t = r\alpha$	